

VIKAS SHARMA

Quantitative Developer | Algorithmic Trading Infrastructure | Python & C++

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PROFESSIONAL SUMMARY

Quantitative Developer with 3+ years architecting enterprise-grade algorithmic trading infrastructure for Indian and US equity and options markets. Built production systems executing **INR 400+ crore (about \$48M) in daily turnover across 150+ accounts**, NumPy-vectorized backtesting engines delivering **150x performance speedups**, and multi-broker SDKs integrating Zerodha, Interactive Brokers, and XTS. Published **pyzdata** on PyPI (Production/Stable). Strong in low-latency Python, distributed messaging (ZeroMQ), and real-time risk systems.

TECHNICAL SKILLS

Languages: Python (Expert; object-oriented design, type hints), C++, SQL, Bash

Quant & Data: NumPy, Polars, Pandas, DuckDB, Numba, Dask, Parquet, Vectorization, Plotly

Systems & Infra: ZeroMQ, WebSocket, AsyncIO, Multithreading, FileLock, Docker, AWS EC2, Linux, Git

Broker APIs: Zerodha Kite, Interactive Brokers (TWS / ib_async), XTS, IBC

Frameworks & Markets: Django, Flask, Streamlit, REST APIs, NSE, BSE, NFO, MCX, CBOE, CME (NIFTY, BANKNIFTY, SENSEX, SPX, SPY, ES)

Quant Concepts: Options & derivatives, synthetic ATM detection, volatility-based (standard deviation) strike routing, portfolio stop-loss design, tick-level time-series processing

PROFESSIONAL EXPERIENCE

Software Engineer II - Pranjali Growcap Pvt. Ltd.

Feb 2024 - Present

Proprietary options trading firm - Indian & US equity / derivatives markets

New Delhi, India

- **Algo Trading SDK:** Architected full-lifecycle trading SDK automating signal generation, multi-broker order execution (Zerodha, IBKR, XTS), real-time RMS, position reconciliation, and P&L tracking across **150+ accounts executing INR 400+ crore (about \$48M) daily turnover**; eliminated 100% of manual order flow.
- **US Options Trading Engine:** Engineered **ZeroMQ hub-spoke architecture** - central TWSMasterHub maintaining a single TWS connection with 5 independent strategy processes; CSV state machines for crash recovery, price-chasing smart order execution, and stale-fill detection; fully autonomous on AWS EC2 from market open to close.
- **Backtesting Engine - 150x Speedup:** Designed NumPy-vectorized engine with $O(\log n)$ price lookup, synthetic future ATM detection, volatility-based (standard deviation) strike routing, and 37-level portfolio stop-loss; reduced backtest cycle from **~150 hours to under 1 hour**.
- **Parallel Backtest Orchestration:** Built FileLock-based job distribution running 16 parallel terminals; processed **50,000+ parameter combos x 250+ dates x 6 DTEs** across NIFTY, BANKNIFTY, SENSEX, MCX, and 40+ US symbols per cycle.
- **Strategy Research:** Backtested **200+ options selling strategies** on 8+ years of tick-level time-series data; built Polars + DuckDB aggregation dashboard with live heatmap visualizations pushed to Excel via xlwings.

Software Engineer - Pranjali Growcap Pvt. Ltd.

Jul 2022 - Jan 2024

Proprietary options trading firm

New Delhi, India

- Built v1 of the firm's proprietary algo trading SDK integrating Zerodha Kite and XTS broker APIs for automated order placement and real-time WebSocket market-data streaming.
- Developed the first in-house backtesting framework for Indian equity and F&O markets, enabling systematic strategy validation and replacing ad-hoc spreadsheet workflows.
- Shipped internal monitoring stack (Streamlit + Telegram bots) for real-time trade alerts, P&L dashboards, and system-health checks used by the trading desk daily.
- Designed data pipelines for cleaning and structuring tick-level market data from NSE and BSE feeds.

PROJECTS

PyZData (pip install pyzdata) - Published on PyPI | *v1.0.6, Production/Stable, MIT License*

- Open-source Python library, CLI, and Streamlit web app for downloading OHLCV + Open Interest data for any NSE, BSE, NFO, or MCX instrument - intervals from 1-minute to daily. Parallel downloads, retries, disk caching, rate-limiting, typed exception hierarchy. **32 GitHub stars, 15 forks, 90 unit tests, PEP 561 typed, trusted publishing via GitHub Actions (Sigstore).**

KiteWeb (28 stars, 9 forks) - Unofficial Python client for Zerodha Kite web endpoints: authentication, session management, and order and market-data calls; adopted across the Indian retail algorithmic-trading community.

Developer Productivity Tools - Jupyter context-menu launcher, PickleParquetReader (double-click .pkl/.parquet to Excel), TerminalSequence (multi-monitor terminal grid), Tele_Easy_Bot (Telegram wrapper).

EDUCATION

BSc Computer Science - Shaheed Sukhdev College of Business Studies, University of Delhi

2020 - 2023

CODING PROFILES

Data structures and algorithms practice: **CodeChef:** 2-star (rating 1495, peak 1559), 252 problems, 10 rated contests **LeetCode:** 65 solved (43 easy, 21 medium, 1 hard) **HackerRank:** SQL Intermediate + 3 verified certs